

Automatic Spectra, Exact Dyadic Cubature, and Probabilistic Duals in the Pascal–Rvachev Hierarchy

A repository-audited frontier report on Fabius, Rvachev, Thue–Morse,
Mahler, q -series, Bell–Bernoulli, and spectral-product structures

Prepared from the ProveIt Fabius-function document corpus

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Abstract

This report audits the current LaTeX research corpus under `Analysis/FabiusFunction/docs` in Vladimir Reshetnikov’s ProveIt repository and develops a set of results not located in the consolidated documents or their manifest descriptions. The audit treats the primary 9509-line exposition, the 14587-line canonical frontier volume, the spectra-and-arithmetic, exponent-and- q -series, dyadic-comb, and frontier-compilation volumes, and the repository’s draft manifest as the principal current representatives of the recursively organized corpus. Historical drafts that have been editorially absorbed are tracked through that manifest. The word “new” below therefore means *new relative to this audited repository state*; it is not an unconditional claim of first publication in the world literature.

The first main result closes a problem explicitly left open in the repository: for every rank $r \geq 1$, the generalized Thue–Morse spectral-sign generating function G_r is nonrational, hypertranscendental over $\mathbb{C}(z)$, and has the unit circle as a natural boundary. The proof combines a new nonperiodicity argument for strongly block-multiplicative digit characters with the Adamczewski–Dreyfus–Hardouin Mahler dichotomy. A parallel construction for the valuation-multiplicity Lambert series

$$L_r(q) = \sum_{n \geq 1} \binom{\nu_2(n) + 1}{r} q^n$$

yields an explicit order- $(r + 1)$ homogeneous Mahler equation, a dense set of radial singularities at dyadic roots of unity, hypertranscendence, and non- D -finiteness. A bivariate master Lambert germ packages all ranks.

The analytic zero divisor then gives a sharp higher-rank Strang–Fix theorem: on the lattice $Q^{-1}\mathbb{Z}$, the density f_r integrates every polynomial of degree strictly below $\binom{\nu_2(Q)+1}{r}$ by an exact finite periodized comb, and the threshold is optimal. Appell reconstruction converts this into exact polynomial reproduction, with immediate Legendre and Bernoulli corollaries.

Wick rotation of the canonical product produces

$$\mathcal{A}_r(w) = \Phi_r(i\sqrt{w}) = \prod_{n \geq 1} \left(1 + \frac{w}{n^2}\right)^{\mu_r(n)}.$$

For every rank this is a type-I Laguerre–Pólya function of order $1/2$; its coefficient sequence is PF_∞ , so all Toeplitz minors, shifted Jensen hyperbolicity statements, and ultra-log-concavity inequalities lift from the known rank-one case to the full Pascal hierarchy.

A new probability interpretation is especially productive. The normalized ratio $\mathcal{A}_r(\theta z)/\mathcal{A}_r(\theta)$ is the probability generating function of an infinite Poisson-binomial spectral count $N_{r,\theta}$. It identifies exact even moments with count probabilities, obeys a rank-branching law, and has a multi-scale limit theory. For $\theta_r = ca^r$, the count is degenerate when $a < 3$, converges in total variation to a fixed Poisson law at $a = 3$, is asymptotically Poisson with diverging mean for $3 < a < \sqrt{15}$, has a nontrivial mod-Poisson correction at $a = \sqrt{15}$, and is asymptotically Gaussian throughout $3 < a < 5$. Finally, $\mathcal{A}_r(w)^{-\tau}$ is the Laplace transform of a generalized-gamma convolution with an

explicit Thorin measure; a separate rank central limit theorem and geometric standardized-cumulant decay follow.

The report supplies complete proofs, exact Bell–Bernoulli formulas, computational checks, a reproducible Python program, a conjecture register, and a staged formalization plan.

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1 Scope, audit method, and originality boundary

1.1 What was audited

The current repository is not a small collection of independent notes. Its own manifest states that absorbed drafts are deleted after their content has been integrated into a primary or consolidated frontier volume, with the manifest and git history serving as the inventory and archive [7]. Accordingly, the audit used two complementary layers.

1. The primary human-readable exposition `Fabius_Function_and_Rvachev_Up.tex`, which separates the bounded Fabius CDF, Rvachev’s compactly supported bump, and the signed global extension, and collects the proved probability, Fourier, arithmetic, inverse, Legendre, and Lambert-asymptotic results [1].
2. The current consolidated frontier documents, especially `semi-formalized-research-frontiers.tex`, `Spectra_and_Arithmetic_Frontiers.tex`, `Exponents_and_q_Series_Frontiers.tex`, `Dyadic_Comb_Frontiers.tex`, and `Frontier_Compilations.tex` [2, 3, 4, 5, 6]. The manifest was then used to identify topics already absorbed from historical or incoming drafts.

This method is stronger than searching a few obvious phrases and weaker than a formal semantic comparison of every theorem in every historical commit. The novelty labels in this report are therefore deliberately local: they record absence from the audited current repository corpus, not an exhaustive global priority search.

1.2 Important duplication exclusions

Several attractive conclusions were initially rediscovered and then removed from the novelty ledger because the corpus already contains them. In particular:

- the canonical product for the classical Rvachev transform, its integer zero multiplicities $1 + \nu_2(n)$, and exact zero jets;
- the Pascal–Rvachev hierarchy, its random-series model, support, refinement law, multiplicity formula $\mu_r(n) = \binom{\nu_2(n)+1}{r}$, spectral zeta function, digital sign law, and Mahler equation [3];
- rank-one sharp dyadic-comb moment exactness and Appell reproduction [5, 6];
- the rank-one type-I Laguerre–Pólya classification, PF_∞ coefficient theorem, Jensen hyperbolicity, moment inequalities, non-infinite-divisibility result, and gamma dual [6];
- higher-rank zero counting, spectral zeta, heat-trace, endpoint, and Fourier-decay results already developed in the spectra volume.

The present results use those statements as inputs. They do not rebrand them as new.

1.3 Result ledger

Topic	Status	Contribution in this report
Spectral signs G_r	new relative to audited repository	Proof of non-eventual-periodicity, nonrationality, hypertranscendence, and the unit-circle natural boundary; this resolves Problem D(iv) stated in the spectra volume.

Topic	Status	Contribution in this report
Multiplicity Lambert series L_r	new relative to audited repository	Divisibility-layer formula, rank recurrence, order- $(r+1)$ Mahler equation, dense radial singularities, natural boundary, hypertranscendence, and non- D -finiteness.
Rank master Lambert germ	new relative to audited repository	The bivariate identity $\mathcal{L}(u, q) = \sum_h (1+u)^h q^{2^h} / (1 - q^{2^h})$ and its first-order Mahler equation.
Higher-rank comb exactness	new relative to audited repository	Sharp degree $\binom{\nu_2(Q)+1}{r} - 1$ for polynomial cubature and Appell reconstruction on $Q^{-1}\mathbb{Z}$.
Laguerre–Pólya/PF hierarchy	new relative to audited repository	Lift of the rank-one product theorem to every Pascal rank, including all Toeplitz-minor and moment inequalities.
Poisson-binomial spectral count	new relative to audited repository	Exact PGF, moment-probability duality, factorial cumulants, rank branching, saddle tilting, Poisson/mod-Poisson/Gaussian phase diagram.
Generalized-gamma dual	new relative to audited repository	All-rank Thorin measure, rank branching, L^2 concentration, and a standardized rank CLT.
Hypertranscendence of Φ_r or $\mathcal{A}_r$	conjectural	Non- D -finiteness is proved; full differential transcendence remains conjectural.
Exact higher-rank dyadic values and inverse asymptotics	Open	Concrete directions are given, but no theorem is claimed here.

Proof-status convention. A theorem or corollary below is proved in the report. A numerical table is supporting evidence only. Every conjecture is marked explicitly. The Python program verifies finite identities and numerical predictions but is not used as a substitute for any analytic proof.

2 Baseline: the Pascal–Rvachev hierarchy

2.1 Fourier convention and random series

Throughout this report

$$\operatorname{sinc}_\pi z := \frac{\sin(\pi z)}{\pi z}, \quad \operatorname{sinc}_\pi 0 = 1.$$

The normalized convention puts the zeros at the nonzero integers.

Definition 2.1 (Pascal–Rvachev products). Set $\Phi_0(z) = \operatorname{sinc}_\pi(2z)$. For $r \geq 1$, let

$$a_{r,h} := \binom{h}{r-1}, \quad \Phi_r(z) := \prod_{h \geq 0} \operatorname{sinc}_\pi\left(\frac{z}{2^h}\right)^{a_{r,h}}. \quad (1)$$

The convention $\binom{h}{k} = 0$ for $h < k$ is always in force.

The repository proves local uniform convergence and gives the random-series model

$$X_r = \sum_{h \geq 0} \sum_{\ell=1}^{a_{r,h}} U_{h,\ell}, \quad U_{h,\ell} \sim \operatorname{Unif}[-2^{-h-1}, 2^{-h-1}], \quad (2)$$

with all summands independent, so that

$$\mathbb{E}e^{-2\pi izX_r} = \Phi_r(z), \quad \text{supp } X_r = [-1, 1]. \quad (3)$$

Write f_r for the density. The case $r = 1$ is the classical Rvachev up-density in the Fourier normalization used here; its translate/integral is the bounded Fabius function.

Pascal’s identity gives the refinement ladder

$$\Phi_r(2z) = \Phi_r(z)\Phi_{r-1}(z), \quad 2X_r \stackrel{d}{=} X_r + X'_{r-1}, \quad (4)$$

where the prime denotes an independent copy.

2.2 Zero divisor and spectral zeta

For $n \neq 0$, the zero order at n is

$$\mu_r(n) = \sum_{h=0}^{\nu_2(|n|)} \binom{h}{r-1} = \binom{\nu_2(|n|)+1}{r}. \quad (5)$$

Consequently

$$\Phi_r(z) = \prod_{n \geq 1} \left(1 - \frac{z^2}{n^2}\right)^{\mu_r(n)}. \quad (6)$$

The associated Dirichlet series is

$$\mathcal{Z}_r(s) := \sum_{n \geq 1} \frac{\mu_r(n)}{n^s} = \zeta(s) \frac{2^{-(r-1)s}}{(1-2^{-s})^r} = \frac{2^s \zeta(s)}{(2^s - 1)^r}, \quad \Re s > 1. \quad (7)$$

These are repository baselines [3].

Two elementary consequences will be used repeatedly:

$$\mathcal{Z}_r(2) = \frac{4\zeta(2)}{3^r}, \quad \mathcal{Z}_r(4) = \frac{16\zeta(4)}{15^r}, \quad \mathcal{Z}_r(2k) = \frac{2^{2k}\zeta(2k)}{(2^{2k} - 1)^r}. \quad (8)$$

Also, the first positive zero is 2^{r-1} , and

$$\mu_r(2^H) = \binom{H+1}{r} \rightarrow \infty. \quad (9)$$

2.3 Moments, cumulants, and Bell polynomials

Let $m_{r,n} = \mathbb{E}X_r^n$. Symmetry kills odd moments. Taking logarithms in (6) gives, for $|z| < 2^{r-1}$,

$$\log \Phi_r(z) = - \sum_{k \geq 1} \frac{\mathcal{Z}_r(2k)}{k} z^{2k}. \quad (10)$$

Equivalently, the even cumulants are

$$\kappa_{2k}(X_r) = (-1)^{k+1} \frac{2(2k-1)!}{(2\pi)^{2k}} \mathcal{Z}_r(2k). \quad (11)$$

This is another repository baseline. We record the Bell-polynomial form because it will connect the analytic and probabilistic results below.

Define

$$\mathcal{A}_r(w) := \Phi_r(i\sqrt{w}) = \sum_{n \geq 0} C_{r,n} w^n. \quad (12)$$

Then

$$C_{r,n} = \frac{1}{n!} \mathcal{B}_n^{\text{exp}}(x_{r,1}, \dots, x_{r,n}), \quad x_{r,k} = (-1)^{k+1} (k-1)! \mathcal{Z}_r(2k), \quad (13)$$

where $\mathcal{B}_n^{\text{exp}}$ is the complete exponential Bell polynomial, and

$$C_{r,n} = \frac{(2\pi)^{2n}}{(2n)!} m_{r,2n}. \quad (14)$$

Euler's formula

$$\zeta(2k) = (-1)^{k+1} \frac{B_{2k} (2\pi)^{2k}}{2(2k)!}$$

makes every $m_{r,2n}$ rational and expresses it explicitly through Bell polynomials in Bernoulli numbers.

3 Hypertranscendence and natural boundaries of the spectral signs

3.1 The generalized Thue–Morse characters

Let $b_j(N) \in \{0, 1\}$ be the binary digits of N . The spectra volume defines

$$\varepsilon_r(N) = (-1)^{\sum_{j \geq 0} b_j(N) \binom{j}{r-1}}, \quad \sigma_{r,j} := (-1)^{\binom{j}{r-1}}. \quad (15)$$

Let

$$P_r := 2^{\lceil \log_2 r \rceil}, \quad B_r := 2^{P_r}. \quad (16)$$

Lucas's theorem implies that $j \mapsto \sigma_{r,j}$ is periodic with period P_r , and at least one entry per period is -1 . The generating function is

$$G_r(z) := \sum_{N \geq 0} \varepsilon_r(N) z^N = \prod_{j \geq 0} (1 + \sigma_{r,j} z^{2^j}). \quad (17)$$

Splitting off one period gives the repository's Mahler equation

$$G_r(z) = A_r^{\text{sign}}(z) G_r(z^{B_r}), \quad A_r^{\text{sign}}(z) = \prod_{j=0}^{P_r-1} (1 + \sigma_{r,j} z^{2^j}). \quad (18)$$

The open frontier list asks for the natural-boundary behavior of G_r on $|z| = 1$ [3].

3.2 A nonperiodicity lemma for periodic digit characters

The key new input is not a numerical observation but an elementary rigidity argument.

Lemma 3.1 (Strong block multiplicativity). *For $0 \leq a < B_r^m$ and $k \geq 0$,*

$$\varepsilon_r(a + B_r^m k) = \varepsilon_r(a) \varepsilon_r(k). \quad (19)$$

Proof. The binary digits of a occupy positions below mP_r , while those of $B_r^m k$ are the digits of k shifted by mP_r . Periodicity of $\sigma_{r,j}$ with period P_r leaves every shifted digit weight unchanged. The two sets of occupied positions are disjoint, so the signs multiply. $\square$

Lemma 3.2 (Eventual periodicity would be pure periodicity). *If ε_r is eventually periodic with period $d \geq 1$, then it is periodic with period d from $N = 0$ onward.*

Proof. Suppose $\varepsilon_r(N + d) = \varepsilon_r(N)$ for $N \geq N_0$. Choose m so large that $B_r^m > N_0 + d$, and then choose k with $B_r^m k \geq N_0$. For $0 \leq a < B_r^m - d$, eventual periodicity and [theorem 3.1](#) give

$$\varepsilon_r(k) \varepsilon_r(a + d) = \varepsilon_r(B_r^m k + a + d) = \varepsilon_r(B_r^m k + a) = \varepsilon_r(k) \varepsilon_r(a).$$

Cancel the sign $\varepsilon_r(k)$. Since m can be arbitrarily large, the identity $\varepsilon_r(a + d) = \varepsilon_r(a)$ holds for every $a \geq 0$. $\square$

Theorem 3.3 (Nonperiodicity of every spectral sign sequence). *For every $r \geq 1$, the sequence $(\varepsilon_r(N))_{N \geq 0}$ is not eventually periodic.*

Proof. Assume it is eventually periodic. By [theorem 3.2](#), it is purely periodic with some period d . Fix j so large that $2^j > d$. Put

$$S_j := \prod_{k=0}^{j-1} \sigma_{r,k}, \quad c := \varepsilon_r(d-1) \in \{\pm 1\}.$$

Pure periodicity gives

$$\sigma_{r,j} = \varepsilon_r(2^j) = \varepsilon_r(2^j - d).$$

In the first j binary positions, $2^j - d = (2^j - 1) - (d - 1)$ is the bitwise complement of $d - 1$. Hence

$$\varepsilon_r(2^j - d) = \prod_{k=0}^{j-1} \sigma_{r,k}^{1-b_k(d-1)} = \frac{S_j}{c}. \quad (20)$$

Therefore $\sigma_{r,j} = S_j/c$, and

$$S_{j+1} = S_j \sigma_{r,j} = \frac{S_j^2}{c} = \frac{1}{c} = c.$$

For all sufficiently large j , both S_j and S_{j+1} equal c , so $\sigma_{r,j} = 1$ eventually. This contradicts the periodic mask's having at least one negative entry in every period. $\square$

3.3 Mahler hypertranscendence

We use the following deep dichotomy of Adamczewski, Dreyfus, and Hardouin [\[8\]](#).

Theorem 3.4 (Mahler differential dichotomy, cited form). *Let $p \geq 2$, and let a possibly ramified Laurent series solve a nontrivial linear Mahler equation over $\mathbb{C}(z)$ involving $f(z), f(z^p), \dots, f(z^{p^m})$. Then either the solution belongs to the ramified rational field $\bigcup_{d \geq 1} \mathbb{C}(z^{1/d})$, or it is hypertranscendental over $\mathbb{C}(z)$.*

A power series in integral powers that lies in a ramified rational field is in fact rational in z : invariance under all deck transformations of a common ramification forces the rational function to descend to $\mathbb{C}(z)$.

Lemma 3.5 (Finite-alphabet rational series). *A rational power series whose coefficients lie in a finite set is eventually periodic.*

Proof. Its coefficients satisfy a constant-coefficient linear recurrence from some index onward. A state consists of a fixed finite window of consecutive coefficients. Only finitely many states occur, and deterministic recurrence propagation makes the state sequence eventually periodic. Therefore the coefficient sequence is eventually periodic. $\square$

Theorem 3.6 (Hypertranscendence and natural boundary of G_r). *For every rank $r \geq 1$:*

- (i) $G_r(z)$ is not rational;
- (ii) $G_r(z)$ is hypertranscendental over $\mathbb{C}(z)$;
- (iii) the unit circle is a natural boundary of G_r .

Thus the natural-boundary part of Problem D in the spectra-and-arithmetic frontier volume has an affirmative answer for every rank.

Proof. The coefficients of G_r lie in $\{\pm 1\}$. If G_r were rational, [theorem 3.5](#) would make ε_r eventually periodic, contradicting [theorem 3.3](#). Equation (18) is a nontrivial linear Mahler equation, so [theorem 3.4](#) and the descent observation imply hypertranscendence.

For the natural boundary there are two independent routes. First, G_r has integer coefficients and radius of convergence exactly 1; the Pólya–Carlson theorem says that such a series is either rational or has the unit circle as a natural boundary. Second, the Mahler-specific theorem of Bell, Coons, and Rowland gives the same rational/natural-boundary dichotomy for a Mahler function meromorphic in the unit disk [9]. Nonrationality rules out the first alternative in either theorem. $\square$

Corollary 3.7 (Generic algebraic independence of values). *Fix $m \geq 0$ and a compact subset K of the open unit disk. For every algebraic point $\alpha \in K$ outside a finite exceptional set, the numbers*

$$G_r(\alpha), G'_r(\alpha), \dots, G_r^{(m)}(\alpha)$$

are algebraically independent over $\overline{\mathbb{Q}}$.

Proof. This is the value-specialization consequence of the Adamczewski–Dreyfus–Hardouin theorem for nonrational Mahler functions [8]. $\square$

Remark 3.8 (What this does not solve). A natural boundary is an analytic-continuation statement. It does not by itself determine rarefied sums, Lyapunov exponents, or the pure-point versus singular-continuous decomposition of the diffraction measure. The other parts of the repository’s Problem D remain open.

3.4 First masks

r	P_r	B_r	negative mask entries	$(\sigma_{r,0}, \dots, \sigma_{r,P_r-1})$
1	1	2		1 —
2	2	4		1 +—
3	4	16		2 ++ --
4	4	16		1 +++—
5	8	256		4 ++++ — — —
6	8	256		2 ++++ + — +—

The Python checks in the archive verify the block identity on large finite ranges for ranks 1 through 8, but [theorem 3.3](#) is the proof.

4 A hypertranscendental hierarchy of valuation Lambert series

4.1 Divisibility layers

Define

$$L_r(q) := \sum_{n \geq 1} \mu_r(n) q^n, \quad |q| < 1. \quad (21)$$

The binomial multiplicity admits a useful layer-cake identity.

Lemma 4.1 (Divisibility decomposition). *For $n \geq 1$,*

$$\mu_r(n) = \sum_{h \geq r-1} \binom{h}{r-1} \mathbf{1}_{2^h | n}. \quad (22)$$

Consequently,

$$L_r(q) = \sum_{h \geq r-1} \binom{h}{r-1} \frac{q^{2^h}}{1 - q^{2^h}}. \quad (23)$$

Proof. If $v = \nu_2(n)$, only $h \leq v$ contribute, and the hockey-stick identity gives $\sum_{h=r-1}^v \binom{h}{r-1} = \binom{v+1}{r}$. Summing $\mathbf{1}_{2^h|n} q^n$ over n gives $q^{2^h} / (1 - q^{2^h})$. $\square$

Formula (23) is a genuine q -series bridge between the Pascal zero divisor and Lambert-series theory. It is distinct from the Dirichlet spectral zeta function (7).

4.2 Rank recursion and a homogeneous Mahler equation

Let T be the Mahler operator $(Tf)(q) = f(q^2)$ and put $R(q) = q/(1 - q)$.

Proposition 4.2 (Pascal–Mahler recursion). *The Lambert hierarchy satisfies*

$$(I - T)L_1 = R, \quad (24)$$

$$(I - T)L_r = TL_{r-1}, \quad r \geq 2, \quad (25)$$

$$(I - T)^r L_r = T^{r-1} R = \frac{q^{2^{r-1}}}{1 - q^{2^{r-1}}}. \quad (26)$$

Proof. Subtract (23) evaluated at q^2 from the same formula at q . Pascal's identity $\binom{h}{r-1} - \binom{h-1}{r-1} = \binom{h-1}{r-2}$ gives (25); the rank-one difference is the first Lambert atom R . Since T commutes with $I - T$, induction gives (26). $\square$

Let $a = 2^{r-1}$. The rational function $g(q) = q^a / (1 - q^a)$ satisfies

$$(1 + q^a)g(q^2) - q^a g(q) = 0.$$

Combining this with (26) gives the promised homogeneous equation.

Theorem 4.3 (Order- $(r + 1)$ Mahler equation). *For every $r \geq 1$,*

$$\boxed{((1 + q^{2^{r-1}})T - q^{2^{r-1}}I)(I - T)^r L_r = 0.} \quad (27)$$

In expanded form this is a nontrivial linear Mahler equation involving $L_r(q), L_r(q^2), \dots, L_r(q^{2^{r+1}})$ with rational coefficients.

4.3 Dense dyadic singularities

Theorem 4.4 (Natural boundary of L_r). *Every dyadic root of unity is a radial singularity of L_r . Since these roots are dense on the unit circle, $|q| = 1$ is a natural boundary.*

Proof. Let ζ have order 2^J and put $q = \rho\zeta$, $0 < \rho < 1$. For every $h \geq \max\{J, r - 1\}$,

$$q^{2^h} = \rho^{2^h} > 0.$$

Thus the tail of (23) is positive real. Already the fixed term with $h = \max\{J, r - 1\}$ tends to $+\infty$ as $\rho \uparrow 1$, while the finitely many preceding terms cannot cancel that divergence. Hence ζ is a radial singularity. An analytic continuation through any open arc of the circle would continue across some dyadic root, a contradiction. $\square$

Theorem 4.5 (Hypertranscendence and nonholonomicity of L_r). *For every $r \geq 1$:*

- (i) L_r is hypertranscendental over $\mathbb{C}(q)$;
- (ii) L_r is not D -finite;
- (iii) the coefficient sequence $\mu_r(n)$ is not P -recursive.

Proof. The natural boundary rules out rationality, including rationality in a ramified variable. Apply [theorem 3.4](#) to (27). Hypertranscendence implies non- D -finiteness. Equivalently, a D -finite ordinary generating function has a P -recursive coefficient sequence, so $\mu_r(n)$ is not P -recursive. $\square$

Remark 4.6 (Regular but not holonomic). For fixed r , $\mu_r(n)$ is nevertheless 2-regular. With the auxiliary convention $\mu_0(n) := 1$,

$$\mu_r(2n) = \mu_r(n) + \mu_{r-1}(n), \quad \mu_r(2n+1) = \begin{cases} 1, & r = 1, \\ 0, & r \geq 2. \end{cases} \quad (28)$$

Thus finite-state linear algebra controls its 2-kernel even though no polynomial-coefficient recurrence in n exists.

4.4 The master rank Lambert germ

Theorem 4.7 (Bivariate rank resolvent). For $|q| < 1$ and u in any bounded set for which the displayed series converges normally,

$$\mathcal{L}(u, q) := \sum_{r \geq 1} u^{r-1} L_r(q) = \sum_{h \geq 0} (1+u)^h \frac{q^{2^h}}{1 - q^{2^h}}. \quad (29)$$

It satisfies the first-order inhomogeneous Mahler equation

$$\boxed{\mathcal{L}(u, q) = \frac{q}{1-q} + (1+u)\mathcal{L}(u, q^2)}. \quad (30)$$

Proof. Interchange the rank and scale sums in (23) and use

$$\sum_{r=1}^{h+1} \binom{h}{r-1} u^{r-1} = (1+u)^h.$$

Separating the $h = 0$ term and shifting $h \mapsto h + 1$ gives (30). $\square$

Remark 4.8 (A q -analogue viewpoint). The factors $q^{2^h}/(1 - q^{2^h})$ are logarithmic derivatives of dyadic q -Pochhammer atoms. The rank parameter changes the scale weight from 1 to $(1+u)^h$. This makes (29) a natural meeting point of automatic sequences, Lambert series, and the repository's broader q -Pochhammer program.

5 Sharp higher-rank dyadic-comb cubature

5.1 A general periodization identity

Let $Q \geq 1$ be an integer lattice parameter; it is denoted Q rather than q to avoid conflict with the Lambert variable. For a polynomial p , put

$$C_{r,Q}[p](x) := \frac{1}{Q} \sum_{k \in \mathbb{Z}} p\left(x - \frac{k}{Q}\right) f_r\left(x - \frac{k}{Q}\right). \quad (31)$$

The sum is finite because f_r is supported on $[-1, 1]$.

Lemma 5.1 (Fourier coefficients of the weighted comb). For $p(t) = t^j$,

$$C_{r,Q}[t^j](x) = \sum_{m \in \mathbb{Z}} \frac{\Phi_r^{(j)}(Qm)}{(-2\pi i)^j} e^{2\pi i Q m x}. \quad (32)$$

The constant coefficient is $m_{r,j}$.

Proof. Apply Poisson summation to $g_j(t) = t^j f_r(t)$ on the lattice $Q^{-1}\mathbb{Z}$. Our Fourier convention gives $\widehat{g_j}_{2\pi}(\xi) = \Phi_r^{(j)}(\xi)/(-2\pi i)^j$. At $\xi = 0$ this is $\int t^j f_r(t) dt = m_{r,j}$. $\square$

5.2 Exact order and sharpness

Theorem 5.2 (Sharp Pascal comb theorem). *Let*

$$M_r(Q) := \mu_r(Q) = \binom{\nu_2(Q) + 1}{r}. \quad (33)$$

If p is a polynomial with $\deg p < M_r(Q)$, then for every real x ,

$$\boxed{\frac{1}{Q} \sum_{k \in \mathbb{Z}} p\left(x - \frac{k}{Q}\right) f_r\left(x - \frac{k}{Q}\right) = \mathbb{E}p(X_r).} \quad (34)$$

The threshold is sharp. If $M_r(Q) > 0$, the identity fails for $p(t) = t^{M_r(Q)}$; if $M_r(Q) = 0$, it already fails for constants.

Proof. For $m \neq 0$,

$$\mu_r(Qm) = \binom{\nu_2(Q) + \nu_2(m) + 1}{r} \geq M_r(Q).$$

Hence every derivative $\Phi_r^{(j)}(Qm)$ vanishes for $j < M_r(Q)$. By linearity and [theorem 5.1](#), all nonconstant Fourier modes disappear for every polynomial of degree below the threshold, leaving the moment at frequency zero.

If $M = M_r(Q) > 0$, the zero at Q has exact order M , so $\Phi_r^{(M)}(Q) \neq 0$. The $m = 1$ Fourier coefficient of the periodization for t^M is therefore nonzero. If $M = 0$, then $\Phi_r(Q) \neq 0$, so even the constant periodization has a nonzero first Fourier mode. $\square$

Corollary 5.3 (Dyadic degree). *For $Q = 2^H$, the exact polynomial degree is*

$$d_{r,H} = \binom{H+1}{r} - 1 \quad (35)$$

when $H \geq r - 1$; when $H < r - 1$, even the constant periodization fails.

For example:

	$H = 2$	$H = 3$	$H = 4$	$H = 5$	$H = 6$	$H = 7$
$r = 1$	2	3	4	5	6	7
$r = 2$	2	5	9	14	20	27
$r = 3$	0	3	9	19	34	55
$r = 4$	–	0	4	14	34	69

Here “–” means $M_r(2^H) = 0$.

Remark 5.4 (Relation to the known rank-one theorem). At $r = 1$, $M_1(Q) = 1 + \nu_2(Q)$, exactly the sharp rank-one hierarchy already proved in the dyadic-comb and frontier-compilation volumes. The new content is the Pascal-rank lift and its binomial growth in $\nu_2(Q)$.

5.3 Appell reconstruction

Let

$$\mathcal{M}_r(z) := \mathbb{E}e^{zX_r} = \Phi_r\left(\frac{iz}{2\pi}\right). \quad (36)$$

It is even. Define Appell polynomials $P_{r,n}$ by

$$\frac{e^{yz}}{\mathcal{M}_r(z)} = \sum_{n \geq 0} P_{r,n}(y) \frac{z^n}{n!}. \quad (37)$$

Theorem 5.5 (Exact polynomial reproduction). *For $0 \leq n < M_r(Q)$,*

$$\boxed{\frac{1}{Q} \sum_{k \in \mathbb{Z}} P_{r,n}\left(\frac{k}{Q}\right) f_r\left(x - \frac{k}{Q}\right) = x^n.} \quad (38)$$

Proof. Apply [theorem 5.2](#) to the polynomial in the variable t

$$p_x(t) = P_{r,n}(x - t).$$

Its degree is $n < M_r(Q)$, so the left side of (38) is $\mathbb{E}P_{r,n}(x - X_r)$. Taking the exponential generating function and using symmetry,

$$\sum_{n \geq 0} \mathbb{E}P_{r,n}(x - X_r) \frac{z^n}{n!} = \frac{e^{xz} \mathbb{E}e^{-zX_r}}{\mathcal{M}_r(z)} = e^{xz}.$$

Coefficient comparison gives x^n . □

5.4 Legendre and Bernoulli corollaries

Let P_n denote the Legendre polynomial and $B_n(x)$ the Bernoulli polynomial.

Corollary 5.6 (Exact finite Legendre moments). *If $n < M_r(Q)$, then for every x ,*

$$\int_{-1}^1 P_n(t) f_r(t) dt = \frac{1}{Q} \sum_{|x-k/Q| \leq 1} P_n\left(x - \frac{k}{Q}\right) f_r\left(x - \frac{k}{Q}\right). \quad (39)$$

Corollary 5.7 (Exact finite Bernoulli moments). *If $n < M_r(Q)$, then*

$$\mathbb{E}B_n(X_r) = \frac{1}{Q} \sum_{|x-k/Q| \leq 1} B_n\left(x - \frac{k}{Q}\right) f_r\left(x - \frac{k}{Q}\right). \quad (40)$$

These identities turn initial blocks of orthogonal and Appell moments into finite exact sampling formulas. They are potentially useful for stable Legendre coefficient computation because the right side is local and the exactness degree is known before any numerical work begins.

5.5 Numerical zero-jet check

The archived program differentiates truncated products at three exact zeros. The first nonzero derivative agrees with $M_r(Q) = \binom{\nu_2(Q)+1}{r}$:

r	Q	predicted order	first nonzero derivative (numerical)
2	4	3	$\Phi_2^{(3)}(4) \approx 1.32282664349 \times 10^{-2}$
3	8	4	$\Phi_3^{(4)}(8) \approx -7.57646125312 \times 10^{-5}$
4	16	5	$\Phi_4^{(5)}(16) \approx 3.51415041684 \times 10^{-8}$

The lower derivatives are numerically below 10^{-130} at the chosen precision. This table checks the implementation; sharpness in [theorem 5.2](#) follows analytically from the exact zero order.

6 All-rank Laguerre–Pólya and Pólya-frequency structure

6.1 Wick rotation

From (6),

$$\mathcal{A}_r(w) := \Phi_r(i\sqrt{w}) = \prod_{n \geq 1} \left(1 + \frac{w}{n^2}\right)^{\mu_r(n)} = \sum_{k \geq 0} C_{r,k} w^k. \quad (41)$$

Since X_r is symmetric,

$$\mathcal{A}_r(w) = \mathbb{E} e^{2\pi\sqrt{w}X_r} = \mathbb{E} e^{-2\pi\sqrt{w}X_r} = \sum_{k \geq 0} \frac{(2\pi)^{2k} m_{r,2k}}{(2k)!} w^k. \quad (42)$$

The square root disappears from the Taylor series.

6.2 Classification and total positivity

Theorem 6.1 (All-rank type-I Laguerre–Pólya theorem). *For every $r \geq 1$, $\mathcal{A}_r$ is a type-I Laguerre–Pólya entire function of genus zero and order $1/2$. Its zeros are exactly $-n^2$, with multiplicity $\mu_r(n)$.*

Proof. The convergence $\sum_n \mu_r(n)/n^2 = \mathcal{Z}_r(2) < \infty$ makes (41) a genus-zero product and the locally uniform limit of polynomials with only nonpositive real zeros. Hence it is type-I Laguerre–Pólya. The exponent of convergence of the zeros is $1/2$ because

$$\sum_{n \geq 1} \frac{\mu_r(n)}{(n^2)^\rho} = \mathcal{Z}_r(2\rho)$$

converges exactly for $\rho > 1/2$. Thus the order is at least $1/2$. On the other hand, $|X_r| \leq 1$ and (42) give growth $\exp(\mathcal{O}(\sqrt{|w|}))$, so the order is at most $1/2$. $\square$

Theorem 6.2 (PF_∞ coefficients and Jensen hyperbolicity). *For every $r \geq 1$, the ordinary coefficient sequence $(C_{r,k})_{k \geq 0}$ is a Pólya-frequency sequence of infinite order. Equivalently, every minor of the Toeplitz matrix $(C_{r,j-i})_{i,j \geq 0}$ is nonnegative. All shifted Jensen polynomials of the exponential coefficient sequence $k!C_{r,k}$ are hyperbolic with nonpositive zeros.*

Proof. The Aissen–Edrei–Schoenberg–Whitney characterization of PF_∞ sequences applies to

$$\mathcal{A}_r(w) = \prod_{n,j} \left(1 + \frac{w}{n^2}\right), \quad 1 \leq j \leq \mu_r(n),$$

because all parameters $1/n^2$ are nonnegative and summable [10]. Jensen’s characterization of the Laguerre–Pólya class then gives the hyperbolicity assertion. $\square$

Corollary 6.3 (Ultra-log-concavity of every rank’s moments). *For $k \geq 1$,*

$$C_{r,k}^2 \geq \frac{k+1}{k} C_{r,k-1} C_{r,k+1}, \quad (43)$$

$$m_{r,2k}^2 \geq \frac{2k-1}{2k+1} m_{r,2k-2} m_{r,2k+2}. \quad (44)$$

Proof. Apply Newton’s inequalities to finite truncations of (41) and pass to the locally uniform limit. Substitution of (14) simplifies the factorial ratio to $(2k-1)/(2k+1)$. $\square$

The numerical ratios

$$\frac{C_{r,k}^2}{((k+1)/k) C_{r,k-1} C_{r,k+1}}$$

are all greater than 1 in the archived table for $1 \leq r \leq 6$ and $1 \leq k \leq 6$. For example, the $k = 1$ ratios are approximately 1.31579, 1.16822, 1.09457, 1.05467, 1.03210 for ranks 1 through 5.

Remark 6.4 (Repository boundary). The rank-one instance, including the PF_∞ theorem and (44), already appears in `Frontier_Compilations.tex`. The theorem here is the all-rank lift through the Pascal multiplicities.

6.3 Rank convolution

For $r \geq 2$, Wick rotation of (4) gives

$$\mathcal{A}_r(4w) = \mathcal{A}_r(w)\mathcal{A}_{r-1}(w). \quad (45)$$

Corollary 6.5 (Coefficient Pascal convolution). *For $r \geq 2$ and every $k \geq 0$,*

$$4^k C_{r,k} = \sum_{j=0}^k C_{r,j} C_{r-1,k-j}. \quad (46)$$

This identity may be read either as a moment convolution or as a positive coefficient recursion inside the PF_∞ cone.

6.4 Non- D -finiteness of the Fourier products

Theorem 6.6 (Unbounded zero orders obstruct holonomicity). *For every $r \geq 1$, neither $\Phi_r(z)$ nor $\mathcal{A}_r(w)$ is D -finite over $\mathbb{C}(z)$.*

Proof. Suppose a nonzero entire function y satisfies a linear differential equation of order d with polynomial coefficients. Away from the finitely many zeros of the leading coefficient, an ordinary-point uniqueness theorem says that a zero of order at least d forces the solution to vanish identically. Thus zero multiplicities at ordinary points are uniformly bounded by $d - 1$.

For Φ_r , the points 2^H have multiplicity $\binom{H+1}{r} \rightarrow \infty$; for $\mathcal{A}_r$, the points -4^H have the same unbounded multiplicities. Infinitely many of those points avoid the finite singular set, a contradiction. $\square$

Conjecture 6.7 (Differential transcendence of the sinc hierarchy). *For every $r \geq 1$, Φ_r and $\mathcal{A}_r$ are hypertranscendental over $\mathbb{C}(z)$. More strongly, finite sets of distinct ranks should be differentially algebraically independent after quotienting by the explicit refinement relations.*

The Mahler dichotomy used for G_r and L_r does not apply directly here: (4) couples different ranks and is multiplicative rather than a scalar linear Mahler equation with rational coefficients.

7 The Poisson-binomial spectral count

7.1 Exact probability generating function

Fix $r \geq 1$ and $\theta > 0$. For every pair (n, j) with $1 \leq j \leq \mu_r(n)$, let $B_{n,j}$ be independent Bernoulli variables with

$$p_{n,\theta} := \mathbb{P}(B_{n,j} = 1) = \frac{\theta}{n^2 + \theta}. \quad (47)$$

Since $\sum_{n,j} p_{n,\theta} \leq \theta \mathcal{Z}_r(2) < \infty$, the sum below is finite almost surely.

Definition 7.1 (Spectral count).

$$N_{r,\theta} := \sum_{n \geq 1} \sum_{j=1}^{\mu_r(n)} B_{n,j}. \quad (48)$$

Theorem 7.2 (Poisson-binomial determinant duality). *The probability generating function of $N_{r,\theta}$ is*

$$\mathbb{E} z^{N_{r,\theta}} = \frac{\mathcal{A}_r(\theta z)}{\mathcal{A}_r(\theta)}. \quad (49)$$

Consequently,

$$\mathbb{P}(N_{r,\theta} = k) = \frac{C_{r,k} \theta^k}{\mathcal{A}_r(\theta)} = \frac{(2\pi)^{2k} m_{r,2k} \theta^k}{(2k)! \mathcal{A}_r(\theta)}. \quad (50)$$

Proof. Each factor in (41) normalizes as

$$\frac{1 + \theta z/n^2}{1 + \theta/n^2} = \frac{n^2}{n^2 + \theta} + \frac{\theta}{n^2 + \theta} z,$$

the PGF of a Bernoulli variable with parameter (47). Multiply the factors with their multiplicities. Coefficient extraction and (14) give (50). $\square$

At $\theta = 1$, the normalized even-moment sequence is literally a probability mass function:

$$\mathbb{P}(N_{r,1} = k) = \frac{(2\pi)^{2k} m_{r,2k}}{(2k)! \mathcal{A}_r(1)}. \quad (51)$$

This is an exact moment–spectrum probability dual, not merely an analogy.

7.2 Cumulants, branching, and coefficient tilting

Proposition 7.3 (Mean, variance, and factorial cumulants). *Let*

$$S_{r,k}(\theta) := \sum_{n \geq 1} \mu_r(n) \left(\frac{\theta}{n^2 + \theta} \right)^k. \quad (52)$$

Then

$$\lambda_{r,\theta} := \mathbb{E} N_{r,\theta} = S_{r,1}(\theta), \quad (53)$$

$$\text{Var}(N_{r,\theta}) = S_{r,1}(\theta) - S_{r,2}(\theta), \quad (54)$$

$$\kappa_k^{(F)}(N_{r,\theta}) = (-1)^{k-1} (k-1)! S_{r,k}(\theta). \quad (55)$$

Proof. Use $\log \mathbb{E}(1+t)^N = \sum_{n,j} \log(1 + p_{n,\theta} t)$ and compare powers of t . $\square$

Theorem 7.4 (Rank branching). *For $r \geq 2$,*

$$N_{r,4\theta} \stackrel{d}{=} N_{r,\theta} + N'_{r-1,\theta}, \quad (56)$$

with independent variables on the right.

Proof. Normalize (45) at $w = \theta$ and at $w = \theta z$; the resulting PGF factorization is exactly convolution of independent counts. $\square$

Equation (50) also gives an exact exponential tilt for coefficient extraction.

Corollary 7.5 (Saddle tilt). *For every $\theta > 0$,*

$$C_{r,k} = \frac{\mathcal{A}_r(\theta)}{\theta^k} \mathbb{P}(N_{r,\theta} = k). \quad (57)$$

The map $\theta \mapsto \lambda_{r,\theta}$ is strictly increasing from 0 to ∞ , so every $k > 0$ has a unique saddle parameter satisfying $\lambda_{r,\theta} = k$.

Proof. The first identity is a rearrangement of (50). Each Bernoulli parameter is strictly increasing in θ , so the mean is strictly increasing. Monotone convergence gives the endpoint limits; as $\theta \rightarrow \infty$, increasingly many spectral levels have probability bounded away from zero. $\square$

This converts asymptotics of the moment coefficients into local limit theory for $N_{r,\theta}$, and conversely.

7.3 Basic comparison inequalities

The following estimates drive the rank phase diagram.

Lemma 7.6 (Zeta comparison). *For every r and $\theta > 0$,*

$$0 \leq \theta \mathcal{Z}_r(2) - \lambda_{r,\theta} \leq \theta^2 \mathcal{Z}_r(4), \quad (58)$$

$$S_{r,2}(\theta) \leq \theta^2 \mathcal{Z}_r(4). \quad (59)$$

Proof. Termwise,

$$0 \leq \frac{\theta}{n^2} - \frac{\theta}{n^2 + \theta} = \frac{\theta^2}{n^2(n^2 + \theta)} \leq \frac{\theta^2}{n^4},$$

and $(\theta/(n^2 + \theta))^2 \leq \theta^2/n^4$. Sum with multiplicities. $\square$

7.4 Critical Poisson limit

Theorem 7.7 (Critical rank Poisson law). *Fix $c > 0$ and put $\theta_r = c3^r$. Then*

$$N_{r,c3^r} \xrightarrow[r \rightarrow \infty]{\text{TV}} \text{Po}(4c\zeta(2)) = \text{Po}\left(\frac{2c\pi^2}{3}\right). \quad (60)$$

With d_{TV} defined as the supremum over events,

$$d_{\text{TV}}(N_{r,c3^r}, \text{Po}(4c\zeta(2))) \leq 48c^2\zeta(4) \left(\frac{3}{5}\right)^r. \quad (61)$$

Proof. By (8), $\theta_r \mathcal{Z}_r(2) = 4c\zeta(2)$. The mean error is at most

$$\theta_r^2 \mathcal{Z}_r(4) = 16c^2\zeta(4)(3/5)^r.$$

Le Cam's inequality for a Poisson-binomial sum gives

$$d_{\text{TV}}(N_{r,\theta_r}, \text{Po}(\lambda_{r,\theta_r})) \leq 2S_{r,2}(\theta_r) \leq 2\theta_r^2 \mathcal{Z}_r(4).$$

The total variation distance between Poisson laws with parameters λ and λ' is at most $|\lambda - \lambda'|$. Add the two bounds. $\square$

For $c = 1$, the limiting mean is $4\zeta(2) = 6.5797362673929057 \dots$. The grouped numerical means are:

r	$\mathbb{E}N_{r,3^r}$	target minus mean	proved TV bound
8	6.343771060790705	2.35965×10^{-1}	8.72586×10^{-1}
12	6.544298628120802	3.54376×10^{-2}	1.13087×10^{-1}
16	6.574932579970129	4.80369×10^{-3}	1.46561×10^{-2}
20	6.579105941567851	6.30326×10^{-4}	1.89943×10^{-3}
24	6.579654307729619	8.19597×10^{-5}	2.46166×10^{-4}
28	6.579725636253052	1.06311×10^{-5}	3.19031×10^{-5}
32	6.579734889287106	1.37811×10^{-6}	4.13465×10^{-6}

7.5 A Poisson, mod-Poisson, and Gaussian phase diagram

The isolated critical result extends to a multi-scale theorem.

Theorem 7.8 (Rank phase diagram). *Fix $c > 0$, $a > 0$, and set $\theta_r = ca^r$. Let $\lambda_r = \mathbb{E}N_{r,\theta_r}$ and $\sigma_r^2 = \text{Var}(N_{r,\theta_r})$.*

- (i) *If $0 < a < 3$, then $N_{r,\theta_r} \rightarrow 0$ in probability and in total variation.*
- (ii) *If $a = 3$, the fixed Poisson limit of [theorem 7.7](#) holds.*
- (iii) *If $3 < a < \sqrt{15}$, then*

$$d_{\text{TV}}(N_{r,\theta_r}, \text{Po}(\lambda_r)) \rightarrow 0, \quad \lambda_r \sim 4c\zeta(2)(a/3)^r \rightarrow \infty. \quad (62)$$

- (iv) *If $a = \sqrt{15}$, then $\lambda_r \rightarrow \infty$ and the following mod-Poisson convergence holds locally uniformly for $t \in \mathbb{C}$:*

$$\mathbb{E}(1+t)^{N_{r,\theta_r}} e^{-\lambda_r t} \rightarrow \exp(-8c^2\zeta(4)t^2). \quad (63)$$

- (v) *Throughout the larger range $3 < a < 5$,*

$$\frac{N_{r,\theta_r} - \lambda_r}{\sigma_r} \xrightarrow{d} \mathcal{N}(0, 1), \quad (64)$$

with

$$\lambda_r \sim \sigma_r^2 \sim 4c\zeta(2)(a/3)^r. \quad (65)$$

Moreover a Berry–Esseen bound is

$$\sup_x \left| \mathbb{P}\left(\frac{N_{r,\theta_r} - \lambda_r}{\sigma_r} \leq x\right) - \Phi_G(x) \right| = \mathcal{O}((3/a)^{r/2}), \quad (66)$$

where Φ_G is the standard normal CDF.

Proof. If $a < 3$, Markov's inequality and $\lambda_r \leq \theta_r \mathcal{Z}_r(2) = 4c\zeta(2)(a/3)^r$ prove (i). Part (ii) was proved above.

For (iii), Le Cam and [\(59\)](#) give

$$d_{\text{TV}}(N_{r,\theta_r}, \text{Po}(\lambda_r)) \leq 32c^2\zeta(4)(a^2/15)^r \rightarrow 0.$$

The mean asymptotic follows from [\(58\)](#), because the relative error is bounded by a constant times $(a/5)^r$.

Now let $a = \sqrt{15}$. The maximum Bernoulli parameter tends to zero because the smallest spectral point is 2^{r-1} and

$$\max_n p_{n,\theta_r} \leq 4c(a/4)^r \rightarrow 0.$$

Furthermore

$$0 \leq \theta_r^2 \mathcal{Z}_r(4) - S_{r,2}(\theta_r) \leq 2\theta_r^3 \mathcal{Z}_r(6) \rightarrow 0,$$

since $15^{3/2}/63 < 1$. Thus $S_{r,2}(\theta_r) \rightarrow 16c^2\zeta(4)$. For every $k \geq 3$,

$$S_{r,k}(\theta_r) \leq \theta_r^k \mathcal{Z}_r(2k) = 2^{2k} c^k \zeta(2k) \left(\frac{15^{k/2}}{4^k - 1} \right)^r \rightarrow 0.$$

Expanding $\log \mathbb{E}(1+t)^N = \sum_{k \geq 1} (-1)^{k-1} S_{r,k} t^k / k$ proves [\(63\)](#); the vanishing maximum parameter makes the remainder uniform on compact t -sets.

Finally suppose $3 < a < 5$. By [theorem 7.6](#),

$$\lambda_r = \theta_r \mathcal{Z}_r(2)(1 + o(1)), \quad S_{r,2}(\theta_r) = o(\lambda_r),$$

so [\(65\)](#) holds and $\sigma_r \rightarrow \infty$. For a Bernoulli variable B with parameter p , $\mathbb{E}|B - p|^3 \leq p(1-p)$. Hence the sum of third absolute centered moments is at most σ_r^2 . The Berry–Esseen theorem, first for finite truncations and then by an L^2 tail limit, gives an error $\mathcal{O}(1/\sigma_r)$, which is [\(66\)](#). $\square$

Remark 7.9 (Two distinct critical scales). The scale 3^r comes from the first spectral trace $\mathcal{Z}_r(2)$ and creates the fixed Poisson law. The scale $(\sqrt{15})^r$ makes the second factorial cumulant order one and creates the mod-Poisson correction. The relative variance correction becomes order one only at the still larger scale 5^r . This hierarchy of critical bases is a direct probabilistic shadow of the spectral denominators $(2^{2k} - 1)^r$.

8 A generalized-gamma spectral dual and its rank CLT

8.1 The Thorin measure

Fix $\tau > 0$. Let

$$V_{r,\tau} := \sum_{n \geq 1} Y_{r,n}, \quad Y_{r,n} \sim \text{Gamma}(\tau \mu_r(n), \text{rate } n^2), \quad (67)$$

with independent summands and zero-shape terms omitted. Since $\sum_n \mathbb{E} Y_{r,n} = \tau \mathcal{Z}_r(2) < \infty$, the sum is finite almost surely.

Theorem 8.1 (Generalized-gamma determinant dual). *For $w \geq 0$,*

$$\boxed{\mathbb{E} e^{-w V_{r,\tau}} = \mathcal{A}_r(w)^{-\tau}}. \quad (68)$$

Thus $V_{r,\tau}$ is a generalized gamma convolution with discrete Thorin measure

$$\mathcal{T}_{r,\tau} = \tau \sum_{n \geq 1} \mu_r(n) \delta_{n^2}. \quad (69)$$

Its cumulants are

$$\kappa_k(V_{r,\tau}) = \tau(k-1)! \mathcal{Z}_r(2k), \quad k \geq 1. \quad (70)$$

Proof. The Laplace transform of a gamma variable with shape α and rate β is $(1 + w/\beta)^{-\alpha}$. Multiply the transforms and use (41). Logarithmic differentiation gives (70). $\square$

Theorem 8.2 (Gamma rank branching). *For $r \geq 2$,*

$$4V_{r,\tau} \stackrel{d}{=} V_{r,\tau} + V'_{r-1,\tau}, \quad (71)$$

with independence on the right.

Proof. Evaluate the Laplace transform of $4V_{r,\tau}$ at w and apply (45). $\square$

8.2 Concentration

From (8),

$$\mathbb{E} V_{r,\tau} = \frac{4\tau \zeta(2)}{3^r}, \quad \text{Var}(V_{r,\tau}) = \frac{16\tau \zeta(4)}{15^r}. \quad (72)$$

Therefore

$$\frac{\text{Var}(V_{r,\tau})}{(\mathbb{E} V_{r,\tau})^2} = \frac{2}{5\tau} \left(\frac{3}{5}\right)^r. \quad (73)$$

Corollary 8.3 (Deterministic mean normalization).

$$\frac{V_{r,\tau}}{\mathbb{E} V_{r,\tau}} \longrightarrow 1 \quad \text{in } L^2. \quad (74)$$

The next result studies the much finer fluctuation scale.

8.3 A rank central limit theorem

Let

$$W_{r,\tau} := \frac{V_{r,\tau} - \mathbb{E}V_{r,\tau}}{\sqrt{\text{Var}(V_{r,\tau})}}. \quad (75)$$

Theorem 8.4 (Generalized-gamma rank CLT). *For every fixed $\tau > 0$,*

$$W_{r,\tau} \xrightarrow{d} \mathcal{N}(0, 1). \quad (76)$$

For $k \geq 3$, its standardized cumulants are exactly

$$\kappa_k(W_{r,\tau}) = \tau^{1-k/2} (k-1)! \frac{\zeta(2k)}{\zeta(4)^{k/2}} \left(\frac{15^{k/2}}{4^k - 1} \right)^r, \quad (77)$$

and hence decay geometrically.

Proof. Formula (77) follows directly from (70) and (72). For every $k \geq 3$, $15^{k/2} < 4^k - 1$, so each fixed standardized cumulant tends to zero.

For a direct characteristic-function proof, put $\sigma_r^2 = \text{Var}(V_{r,\tau})$. Then

$$\log \mathbb{E} e^{itW_{r,\tau}} = -\frac{it\mathbb{E}V_{r,\tau}}{\sigma_r} - \tau \sum_n \mu_r(n) \log \left(1 - \frac{it}{\sigma_r n^2} \right).$$

The smallest active n is 2^{r-1} , and

$$\sigma_r 2^{2r-2} = \sqrt{\tau \zeta(4)} \left(\frac{4}{\sqrt{15}} \right)^r \rightarrow \infty. \quad (78)$$

Thus the logarithm may be expanded uniformly for fixed t . The linear term cancels the centering, and the quadratic term is $-t^2/2$. The absolute remainder is bounded by a constant multiple of

$$|t|^3 \frac{\tau \mathcal{Z}_r(6)}{\sigma_r^3} = \mathcal{O} \left(\left(\frac{15^{3/2}}{63} \right)^r \right) \rightarrow 0.$$

Lévy's continuity theorem proves the CLT. □

The numerical standardized cumulants for $\tau = 1$ are:

r	κ_3	κ_4	κ_5	κ_6
8	9.4478×10^{-1}	1.8895	5.4648	2.0155×10^1
16	4.9397×10^{-1}	6.9420×10^{-1}	1.5149	4.2908
24	2.5827×10^{-1}	2.5505×10^{-1}	4.1996×10^{-1}	9.1347×10^{-1}
32	1.3503×10^{-1}	9.3704×10^{-2}	1.1642×10^{-1}	1.9447×10^{-1}
48	3.6914×10^{-2}	1.2648×10^{-2}	8.9469×10^{-3}	8.8140×10^{-3}
64	1.0091×10^{-2}	1.7073×10^{-3}	6.8756×10^{-4}	3.9947×10^{-4}

9 Connections to Bell, Bernoulli, Legendre, q -series, and inverse problems

9.1 Exact Bell–Bernoulli moment calculus

Combining (13) and (14) gives the explicit closed formula

$$m_{r,2n} = \frac{(2n)!}{(2\pi)^{2n} n!} \mathcal{B}_n^{\text{exp}}(\mathcal{Z}_r(2), -1! \mathcal{Z}_r(4), \dots, (-1)^{n+1} (n-1)! \mathcal{Z}_r(2n)). \quad (79)$$

Substitution of Euler’s zeta values converts this to a rational expression in $B_2, B_4, \dots, B_{2n}$ and the denominators $(2^{2k} - 1)^r$. The first exact moments, also reproduced by the Python program, include

$$\begin{array}{lll} m_{1,2} = \frac{1}{9}, & m_{1,4} = \frac{19}{675}, & m_{1,6} = \frac{583}{59535}, \\ m_{2,2} = \frac{1}{27}, & m_{2,4} = \frac{107}{30375}, & m_{2,6} = \frac{27953}{56260575}, \\ m_{3,2} = \frac{1}{81}, & m_{3,4} = \frac{571}{1366875}, & m_{3,6} = \frac{1165663}{53166243375}. \end{array}$$

The new PF_∞ theorem adds a family of inequalities to this exact arithmetic data. Every Toeplitz minor becomes a polynomial inequality among these Bell–Bernoulli expressions. This is stronger than the first quadratic Turán inequality and gives a systematic target for symbolic verification.

9.2 Legendre data from exact combs

The repository already gives a Fourier–Legendre expansion of the classical up-function. The higher-rank comb theorem contributes a complementary local route. For $n < M_r(Q)$, the Legendre coefficient

$$\ell_{r,n} := \frac{2n+1}{2} \int_{-1}^1 f_r(t) P_n(t) dt \quad (80)$$

can be computed from any phase x by the finite sum in (39). Varying x provides exact internal consistency checks. Choosing $Q = 2^H$ makes the exactness degree grow like $H^r/r!$, a dramatic higher-rank gain over the linear rank-one degree.

A promising computational program is to combine this exact low-degree block with the repository’s global Legendre tail estimates. The comb handles the initial coefficients without quadrature error; only the tail needs analytic bounding.

9.3 Mahler and q -series bridges

There are now two different Mahler mechanisms in the same hierarchy.

- The sign functions G_r are products of binary cyclotomic-like factors and satisfy a first-order homogeneous Mahler equation at the block base $B_r = 2^{Pr}$.
- The multiplicity series L_r are weighted dyadic Lambert series. Their rank recursion produces an order- $(r+1)$ homogeneous equation at base 2.

Both are hypertranscendental, but for different structural reasons: G_r needs a digit-character nonperiodicity argument, while L_r has a direct dense set of radial singularities. The master germ (29) adds a rank variable to the q -series side and may be viewed as a resolvent of the scale-shift operator.

9.4 Inverse Fabius and Lambert- W directions

The primary repository exposition already develops endpoint and inverse asymptotics with exact lower-Lambert coordinates and periodic corrections. The results here suggest three nonduplicative continuations.

First, let $Q_r(u)$ be the quantile function of X_r . Then

$$\mathbb{E}p(X_r) = \int_0^1 p(Q_r(u)) du.$$

Combining this with (34) transports every polynomial below the sharp threshold into an exact identity between a quantile integral and a finite density comb. For $r = 1$, translating between X_1 , the up-density, and the bounded Fabius CDF turns this into exact integral constraints on the inverse Fabius function.

Second, the saddle identity (57) reduces coefficient asymptotics for $C_{r,k}$ to inversion of $\lambda_{r,\theta} = k$. In joint large- (r, k) regimes, the dyadic pole lattice of the spectral zeta should produce logarithmically periodic corrections. Determining whether the nonlinear inversion is most naturally expressed in a Lambert- W coordinate is an open problem rather than a claim of this report.

Third, the mod-Poisson scale $\theta_r = c(\sqrt{15})^r$ is governed by the second spectral trace, whereas endpoint inversion is governed by Mellin poles at the imaginary dyadic lattice. A two-parameter saddle analysis could reveal a common periodic phase connecting these apparently different Lambert phenomena.

10 Conjectures and frontier research program

10.1 Differential algebra

Conjecture 10.1 (Hypertranscendence of the Fourier hierarchy). *For each $r \geq 1$, the entire functions Φ_r and $\mathcal{A}_r$ are hypertranscendental over $\mathbb{C}(z)$.*

Conjecture 10.2 (Cross-rank differential independence). *After adjoining the explicit refinement relations $\Phi_r(2z) = \Phi_r(z)\Phi_{r-1}(z)$, no further algebraic differential relations exist among a finite set of ranks.*

A possible route is parametrized difference Galois theory for the triangular rank system rather than for a single scalar Mahler equation.

10.2 Strict total positivity and zero interlacing

Conjecture 10.3 (Strictness). *After removing the forced degeneracies caused by repeated spectral zeros, all admissible Toeplitz minors of $(C_{r,k})$ are strictly positive, and the zeros of consecutive nontrivial Jensen polynomials strictly interlace.*

Finite products reduce this to strict total positivity for elementary symmetric functions with repeated parameters. The difficulty is formulating the exact admissibility condition that survives the infinite limit.

10.3 Beyond the proved Poisson-binomial phase range

Theorem 7.8 identifies the bases 3, $\sqrt{15}$, and 5. Several further transitions should occur.

Problem 10.4 (The 5^r variance-critical law). Determine the limiting centered law when $\theta_r = c5^r$. At this scale the second Bernoulli power sum is of the same order as the mean, so the variance is no longer asymptotic to the mean. Higher factorial cumulants should decide whether the correct limit is still Gaussian after a modified normalization.

Problem 10.5 (Approach to the local saturation scale 4^r). The largest Bernoulli parameter changes from vanishing to order one near $\theta_r \asymp 4^r$, even though the global CLT survives beyond that scale. Separate the finitely many saturated low levels from the diffuse high-level cloud and derive a convolution limit.

Conjecture 10.6 (Full mod-Poisson hierarchy). *For each $k \geq 2$, the critical base $a_k = (4^k - 1)^{1/k}$ produces a mod-Poisson correction whose first surviving term is the k th factorial cumulant. The bases increase to 4, creating an accumulating hierarchy of local spectral transitions.*

The statement needs care because lower factorial cumulants may already diverge; the correct renormalization should subtract all preceding terms.

10.4 Gamma Edgeworth theory

Problem 10.7 (Explicit Edgeworth expansion). Use (77) to derive an all-orders Edgeworth expansion for $W_{r,\tau}$ with a remainder uniform in a growing central window. The smallest geometric decay factor is $15^{3/2}/63$, so the skewness term should control the first correction.

The characteristic-function proof already gives an expanding analytic disk of radius proportional to $(4/\sqrt{15})^r$, an unusually favorable setting for uniform inversion.

10.5 Exact arithmetic and formalization

Problem 10.8 (Denominator geometry). Combine the Bell formula (79), von Staudt–Clausen, and the rank denominators $(2^{2k} - 1)^r$ to determine exact prime valuations of $m_{r,2n}$. The PF_∞ inequalities may impose additional divisibility or sign constraints on reduced numerators.

Problem 10.9 (Machine-checked Mahler frontier). Formalize the nonperiodicity theorem, the Lambert operator recurrence, and the sharp comb theorem. The analytic imports needed for full hypertranscendence are substantial, but most of the new finite combinatorics and exact Fourier zero-order reasoning can be separated from them.

11 Formalization roadmap

A realistic Lean development can be staged so that each layer has independent value.

11.1 Tier I: finite combinatorics

1. Define $\sigma_{r,j}$ and $\varepsilon_r(N)$ from binary digits.
 2. Prove periodicity of the mask by Lucas’s theorem.
 3. Prove strong B_r -multiplicativity.
 4. Formalize [theorem 3.2](#) and the complement-bit identity (20).
 5. Conclude non-eventual-periodicity.
 6. Prove the layer-cake identity (22) and the coefficient recurrences (24)–(26) in formal power series.
- These steps require no complex analysis.

11.2 Tier II: exact products and Fourier algebra

1. Reuse the repository’s weighted-scale multiplicity theorem to identify zero orders.
2. Formalize the periodized Fourier coefficient calculation (32) under the existing Poisson-summation API.
3. Prove the exactness and sharpness theorem 5.2.
4. Define the Appell family and prove (38) by finite polynomial algebra plus moment evaluation.
5. Formalize the product identity for $\mathcal{A}_r$ and coefficient convolution (46).

11.3 Tier III: probability

1. Construct countable independent Bernoulli variables with summable parameters and identify the PGF (49).
2. Prove the mean and power-sum bounds in [theorem 7.6](#).

3. Import or prove a countable version of Le Cam’s inequality by finite truncation.
4. Establish the critical Poisson theorem and the subcritical regime.
5. Formalize Berry–Esseen for finite truncations and pass to the limit.
6. Construct the gamma convolution from summable means and prove its Laplace transform and cumulants.

11.4 Tier IV: external transcendence theorems

The final hypertranscendence statements depend on the deep external theorem 3.4. A practical formal boundary would encode that theorem as an explicit imported assumption, prove that G_r and L_r satisfy its hypotheses, and keep the elementary nonrationality and natural-boundary proofs independent.

12 Computational experiments and reproducibility

12.1 What the program checks

The archive contains `frontier_experiments.py`. It performs the following deterministic checks at 80 decimal digits.

1. Coefficientwise verification of $(I - T)^r L_r = T^{r-1} q / (1 - q)$ through rank 8 and degree 4096.
2. Block multiplicativity of ε_r through rank 8 and large finite blocks, plus a finite search excluding small periods.
3. Exact rational moments from Bernoulli numbers and the Bell recurrence.
4. PF_∞ ultra-log-concavity ratios for ranks 1 through 6.
5. Numerical zero jets at $(r, Q) = (2, 4), (3, 8), (4, 16)$.
6. Grouped Poisson-binomial means at $\theta = 3^r$, with rigorous analytic error bounds.
7. Standardized gamma cumulants through rank 64.
8. Radial growth of L_r toward an eighth root of unity.

The exact command is

```
python frontier_experiments.py --output-dir numerical_output --dps 80
```

It creates `numerical_results.txt`, `moment_ulc.csv`, `poisson_limit.csv`, `gamma_clt.csv`, and `lambert_radial.csv`.

12.2 Interpretive limits

The finite sign search does not prove nonperiodicity; theorem 3.3 does. The radial table does not prove a natural boundary; density of the proved dyadic radial singularities does. The small numerical values of lower zero derivatives reflect truncation and finite precision; their exact vanishing follows from the multiplicity theorem.

13 Conclusion

The repository’s existing work already connects the Fabius and Rvachev functions to probability, exact dyadic arithmetic, infinite sinc products, Lambert asymptotics, Legendre expansions, Bell–Bernoulli moment calculus, q -binomial identities, and a higher Pascal hierarchy. The most productive way to

advance that corpus was therefore not to derive another version of the same endpoint or zero-count formula, but to ask what the established divisor and automatic sign structure force in neighboring theories.

That strategy produced five coherent advances.

First, the spectral signs are not merely automatic: their generating functions are hypertranscendental and analytically blocked by the unit circle. Second, the zero multiplicities have their own Lambert–Mahler hierarchy with the same strong analytic obstruction. Third, exact lattice integration grows from linear order in the classical rank to binomial order in the Pascal rank. Fourth, Wick rotation promotes the entire zero divisor to total positivity at every rank. Fifth, the same product is a probability generating function and a Laplace transform, yielding two new spectral probability models with exact branching laws and rank limit theorems.

These results suggest a broader principle: the dyadic zero divisor is not just a bookkeeping device for an infinite product. Under different transforms it becomes an automatic character, a Lambert q -series, a Strang–Fix exactness order, a Pólya-frequency sequence, a Poisson-binomial cloud, and a Thorin measure. Future progress is likely to come from moving deliberately between those representations rather than treating them as separate topics.

A Auxiliary inequalities for the Poisson phase diagram

For reference, the elementary estimates used in [theorem 7.8](#) can be organized by Bernoulli power sums. With $p_n = \theta/(n^2 + \theta)$ and multiplicities understood,

$$S_k(\theta) = \sum \mu_r(n) p_n^k \leq \theta^k \mathcal{Z}_r(2k). \quad (81)$$

Moreover, for $k \geq 1$,

$$0 \leq \theta^k \mathcal{Z}_r(2k) - S_k(\theta) \leq k \theta^{k+1} \mathcal{Z}_r(2k+2), \quad (82)$$

because $1 - (1+x)^{-k} \leq kx$ for $x \geq 0$. At a critical scaling $\theta_r = ca^r$, the leading upper bound is

$$\theta_r^k \mathcal{Z}_r(2k) = 2^{2k} c^k \zeta(2k) \left(\frac{a^k}{4^k - 1} \right)^r. \quad (83)$$

This one formula explains the sequence of factorial-cumulant critical bases $(4^k - 1)^{1/k}$.

B Exact rational moment tables

The following values were generated symbolically from (79); no floating-point reconstruction was used.

r	$m_{r,0}$	$m_{r,2}$	$m_{r,4}$
1	1	1/9	19/675
2	1	1/27	107/30375
3	1	1/81	571/1366875
4	1	1/243	2963/61509375

r	$m_{r,6}$	$m_{r,8}$
1	583/59535	132809/32531625
2	27953/56260575	698600719/7839308334375
3	1165663/53166243375	2864068312649/1889077325876015625
4	45163673/50242099989375	10285851106584079/455220408602972865234375

C Repository and literature references

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